

Def. Let $\beta: I \rightarrow \mathbb{R}^3$ be a unit-speed curve.

A vector field β' is called the unit tangent vector field on β , denote $T = \beta'$.

The derivative $T' = \beta''$ is called the curvature vector field of β .

The value $k(s) = \|T'(s)\|$ ($s \in I$) is called the curvature function of β .

To restriction s.t. $k > 0$,
the unit-vector field $N = \frac{T'}{k}$ is called the principal normal vector field of β .

The cross product $B = T \times N$ on β is called the binormal vector field of β .

Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $k > 0$,

Then, the three vector fields T, N , and B on β are unit vector fields that are mutually orthogonal at each point.

Proof. By definition, $\|T\| = \|\beta'\| = 1$. And since the curvature $k > 0$, $\|N\| = \frac{1}{k} \cdot \|T'\| = 1$.

And, since $\|T\|^2 = T \cdot T$ is constant, $2T \cdot T' = 0$, so $T \perp N$.

therefore the binormal, $\|B\| = \|T \times N\|^2 = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$.

In summary, $T = \beta'$, $N = T'/k$, $B = T \times N$, and $\{T, N, B\}$ is orthonormal.

Since $B \cdot T = 0$, $B' \cdot T + B \cdot T' = 0$ by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot kN = 0$$

And since B is unit, $B' \cdot B = 0$, thus, there is a scalar value function τ s.t.

$$B' = -\tau N$$

, called the torsion. Now

Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $k > 0$ and torsion τ ,

$$\begin{aligned} T' &= kN \\ N' &= -kT + \tau B \\ B' &= -\tau N \end{aligned} \quad \begin{bmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof. The first and third rows are just definition. To show the second, observe $N' = (N' \cdot T)T + (N' \cdot N)N + (N' \cdot B)B$

Now, $N' \cdot T = N \cdot T' = 0$ and $N' \cdot B = N \cdot B' = 0$ gives the results.

Prop. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$. Then,

β is a plane curve if and only if $\tau = 0$, the torsion is zero.

Proof. Suppose that β is a plane curve. That, $\text{Im } \beta$ lies in some plane.

Therefore, there are two vectors $p, q \in \mathbb{R}^3$ such that $q \neq 0$ and

$$[\beta(s) - p] \cdot q = 0 \text{ for all } s.$$

Differentiating yields:

$$\beta'(s) \cdot q = 0 \text{ and } \beta''(s) \cdot q = 0 \text{ for all } s.$$

That is, q is orthogonal to $T = \beta'$ and $N = \beta''/\kappa$. So, β and q are parallel.

We can conclude that $\beta = \pm q/||q||$. So $\beta' = 0$, the torsion τ must be zero.

Conversely, suppose that the torsion is zero. Then, $\beta' = 0$ by definition.

($\beta' = 0$ means β is parallel.) To show that $\text{Im } \beta$ lies in some plane, define

$$f(s) = [\beta(s) - \beta(0)] \cdot \beta(s) \text{ for all } s.$$

Then, the derivative is

$$f'(s) = \beta'(s) \cdot \beta(s) + \beta(s) \cdot \beta'(s) \stackrel{\text{def.}}{=} \beta'(s) \cdot \beta(s) = T \cdot \beta = 0.$$

By definition, $f(0) = 0$, so $f = 0$. Consequently,

$$[\beta(s) - \beta(0)] \cdot \beta = 0 \text{ for all } s,$$

So proved $\square$

Prop. Let β be a unit speed curve with constant curvature $\kappa > 0$ and torsion zero.

Then, the β is part of a circle of radius $1/\kappa$.

Proof. Since $\tau = 0$, the β is a plane curve. Define a curve

$$\gamma(s) = \beta(s) + \frac{1}{\kappa} \cdot N(s).$$

By the Frenet formulae, $\gamma' = \beta' + \frac{1}{\kappa} \cdot N' = T + \frac{1}{\kappa} \cdot (-\kappa T) = 0$. Therefore, γ is constant.

Now, $d(\gamma, \beta(s)) = \|\gamma - \beta(s)\| = \frac{1}{\kappa} \cdot \|N\| = \frac{1}{\kappa}$, so proved.